

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) A manufacturer claims a population mean accuracy of 0.95. Which null hypothesis tests this claim? (1)
(A) $\mu = 0.95$
(B) $\bar{x} = 0.95$
(C) $\mu \neq 0.95$
(D) $\mu < 0.95$
- 2) A population has known standard deviation 0.02. For an independent sample of size 36, what is the standard error of the sample mean? (1)
(A) 0.12
(B) 0.02
(C) $0.02/6$
(D) $0.02/36$
- 3) With $\bar{x} = 0.947$, $\mu_0 = 0.95$, $\sigma = 0.02$ and $n = 36$, what is the z statistic? (1)
(A) 0.90
(B) -0.90
(C) -9.00
(D) 9.00
- 4) Observations range from 180 to 480. What is the width of each of three equal-width bins? (1)
(A) 60
(B) 150
(C) 300
(D) 100
- 5) Twelve observations are divided into three equal-frequency bins. How many observations are in each bin? (1)
(A) 3
(B) 4
(C) 6
(D) 12

Section B: Theory questions (25 marks)

- 6) (i) Find the standard deviation and skewness for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Give an example of the combining-variables concept in dataset preparation.
(iii) A CMO compares three new cola flavours using ratings from 12 customers on a 1-9 scale. Conduct the appropriate analysis. Ratings: Flavour A: 4, 2, 4, 2; Flavour B: 6, 4, 5, 5; Flavour C: 8, 6, 8, 6. Use population standard deviation and Pearson's second skewness coefficient. For (iii), assume independent normal samples with equal variances; use a one-way ANOVA at 5%, with critical $F(2,9) = 4.256$.
- 7) (i) Find the elements of a box plot for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Why is it sometimes necessary to create subsets from a large dataset before analysis?
(iii) Measure the strength of the relationship between the rankings assigned by two interviewers to candidates A-L. Interviewer 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Interviewer 2: 1, 2, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11. For quartiles, use medians of the two halves. Use Spearman rank correlation in (iii).
- 8) A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time. A researcher believes the true accuracy is less than 95%. A sample of 50 test results has mean accuracy 0.93; the known population standard deviation is 0.04. Test the claim at the 5% significance level. Critical z-value: -1.645. (3)
- 9) Test the effectiveness of vaccination in preventing smallpox at the 5% significance level. (5)
Vaccinated: attacked 31, not attacked 469, total 500.
Not vaccinated: attacked 185, not attacked 1315, total 1500.
Overall: attacked 216, not attacked 1784, total 2000. Use Pearson chi-square without continuity correction; critical value for 1 degree of freedom is 3.841. State association, not a causal conclusion.
- 10) Per-acre wheat production (tons) is recorded for three varieties and four fertilizers, with each combination replicated twice. Test whether variety differences and fertilizer differences are significant at the 5% level. Each pair below gives the two replicates, in fertilizer order F1, F2, F3, F4. (5)
Variety 1: (8,9), (9,8), (6,7), (10,11).
Variety 2: (7,6), (8,9), (5,4), (9,10).
Variety 3: (10,9), (11,12), (8,7), (12,13). Assume independent normal errors with common variance. Include the interaction term. Use critical F values: variety (2,12): 3.885; fertilizer (3,12): 3.490.
- 11) A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why. (2)

End of question paper

Worked Solutions

Data Analytics

Set A: Ensemble | CSE Semester 3

Companion to da-ensemble.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Null hypothesis

Correct option: (A). The null hypothesis specifies the population parameter: $H_0 : \mu = 0.95$.

2. Standard error

Correct option: (C). $SE = \sigma / \sqrt{n} = 0.02 / \sqrt{36} = 0.02/6$.

3. Normal-test statistic

Correct option: (B). $z = (0.947 - 0.95) / (0.02/6) = -0.90$.

4. Equal-width bins

Correct option: (D). Bin width is $(480 - 180) / 3 = 100$.

5. Equal-frequency bins

Correct option: (B). Each bin contains $12/3 = 4$ observations.

6. Descriptive statistics, preparation and ANOVA

(i) $N = 20$, $\sum x_i = 1350$, $\bar{x} = 67.5$, and the median is 63. Using the population denominator,

$$\sigma^2 = \frac{\sum (x_i - 67.5)^2}{20} = 238.35, \quad \sigma = 15.4386.$$

Pearson's second coefficient is $3(67.5 - 63) / 15.4386 = 0.8744$, indicating positive skewness.

(ii) Combine quantity and unit price to create revenue: $\text{revenue} = \text{quantity} \times \text{unit price}$. This gives a useful derived variable from two existing columns.

(iii) Test $H_0 : \mu_A = \mu_B = \mu_C$ against at least one unequal mean. The group means are 3, 5 and 7; the grand mean is 5.

$$SS_B = 4[(3 - 5)^2 + (5 - 5)^2 + (7 - 5)^2] = 32, \quad SS_W = 4 + 2 + 4 = 10.$$

Source	SS	df	MS	F
Between	32	2	16	14.4
Within	10	9	1.1111	
Total	42	11		

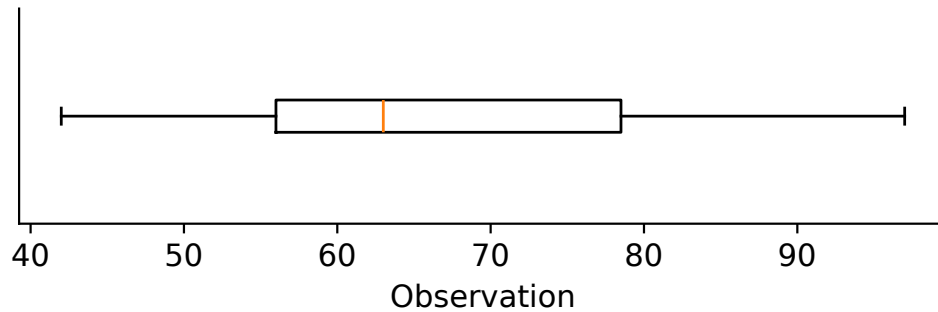
Since $14.4 > 4.256$, reject H_0 at 5%. Mean flavour ratings differ. The test alone does not identify which pairs differ.

7. Box plot and Spearman correlation

(i) Sorted observations:

42, 47, 50, 50, 54, 58, 60, 61, 62, 63, 63, 68, 75, 78, 78, 79, 82, 91, 92, 97.

The five-number summary is $[42, 56, 63, 78.5, 97]$. Here $Q_1 = (54 + 58)/2 = 56$, $Q_3 = (78 + 79)/2 = 78.5$, and $IQR = 22.5$. The fences are 22.25 and 112.25, so there are no outliers.



(ii) Subsets support analysis of relevant cohorts, reduce computational load, and allow separate training and validation sets. Select them appropriately to avoid introducing bias.

(iii) The first two ranks agree. The other ten items have rank differences ± 1 , so $\sum d_i^2 = 10$.

$$r_s = 1 - \frac{6(10)}{12(12^2 - 1)} = \boxed{0.9650}.$$

The interviewers have strong positive rank agreement.

8. Left-tailed normal test

Under the supplied model, $H_0 : \mu = 0.95$ and $H_1 : \mu < 0.95$.

$$z = \frac{0.93 - 0.95}{0.04/\sqrt{50}} = \boxed{-3.5355}.$$

Since $-3.5355 < -1.645$, reject H_0 . The evidence supports a mean accuracy below 95% under this model ($p \approx 0.000203$). The supplied standard deviation is treated as that of accuracy measurements, not inferred as a Bernoulli standard deviation from individual positive/negative results.

9. Vaccination and disease: chi-square test

H_0 : vaccination status and disease status are independent. H_1 : they are associated. Compute each expected count as row total times column total divided by 2000.

Group	Attacked: observed / expected	Not attacked: observed / expected
Vaccinated	31 / 54	469 / 446
Not vaccinated	185 / 162	1315 / 1338

$$\chi^2 = \frac{23^2}{54} + \frac{23^2}{446} + \frac{23^2}{162} + \frac{23^2}{1338} = \boxed{14.6432}.$$

There is one degree of freedom. Since $14.6432 > 3.841$, reject independence. The observed attack rates are $31/500 = 6.2\%$ and $185/1500 = 12.33\%$. Vaccination is associated with a lower attack rate in this sample; the table alone does not establish causation.

10. Two-factor ANOVA with replication

There are $a = 3$ varieties, $b = 4$ fertilizers and $r = 2$ replicates per cell. The grand mean is 8.6667. Variety means are (8.5, 7.25, 10.25) and fertilizer means are (8.1667, 9.5, 6.1667, 10.8333). Use

$$SS_A = br \sum_i (\bar{y}_{i..} - \bar{y})^2, \quad SS_B = ar \sum_j (\bar{y}_{.j.} - \bar{y})^2,$$

$$SS_{AB} = r \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y})^2,$$

$$SS_E = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2.$$

Source	SS	df	MS	F
Variety	36.3333	2	18.1667	36.3333
Fertilizer	71.3333	3	23.7778	47.5556
Interaction	3.6667	6	0.6111	1.2222
Error	6	12	0.5	
Total	117.3333	23		

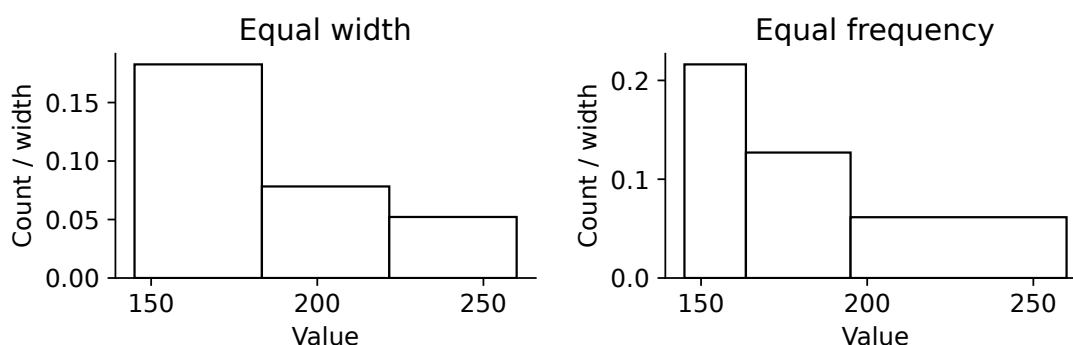
Reject equal variety means because $36.3333 > 3.885$. Reject equal fertilizer means because $47.5556 > 3.490$. Both factors have significant effects at 5% under the supplied assumptions. The interaction is retained in the model; its statistic is shown for completeness. The residual error mean square, rather than the interaction mean square, is the denominator for these fixed-factor tests.

11. Cholesterol binning

Range = $260 - 145 = 115$, so width = $115/3 = 38.3333$.

Method	Bin 1	Bin 2	Bin 3
Equal width	[145,183.333): 7	[183.333,221.667): 3	[221.667,260]: 2
Equal frequency	145,155,160,162	165,172,180,190	200,220,240,260

For equal frequency, use boundaries 145, 163.5, 195 and 260. Equal width better displays the declining concentration across these data and the upper tail. Equal frequency gives balanced groups but requires density heights to make its unequal-width histogram meaningful.



MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which data structure directly supports undoing the most recent action first? (1)
(A) Queue
(B) Stack
(C) Sorted array
(D) Adjacency matrix
- 2) Edits A, B and C are pushed in that order onto an undo stack. Which edit does the first undo remove? (1)
(A) A
(B) B
(C) All three
(D) C
- 3) After edits A, B and C, two undo operations transfer actions to a redo stack. What is the redo stack from bottom to top? (1)
(A) C, B
(B) B, C
(C) A, B
(D) A, C
- 4) The redo stack contains C, B from bottom to top. Which edit is reapplied by one redo operation? (1)
(A) A
(B) C
(C) B
(D) Both B and C
- 5) In a conventional two-stack editor, what happens to the redo stack when a new edit is made after an undo? (1)
(A) It is reversed
(B) It is cleared
(C) It is unchanged
(D) It becomes the undo stack

Section B: Theory questions (25 marks)

- 6) Write a C program to read and display employee details including the date of joining using nested structure. Consider date of joining as nested structure. (2)
- 7) Assume that you are storing marks of students in 3 subjects. Write a C program that: i. Takes the number of students as input. ii. Dynamically allocates a 2D array using pointers. iii. Accepts marks for each subject. Calculates and prints the average marks of each student using only pointer arithmetic. (2)
- 8) Write a C program to dynamically allocate memory for 2D string arrays (array of strings) and perform concatenation using pointers. (2)
- 9) A music streaming service keeps track of songs in a playlist using a singly linked list. Each node stores the following details: SongID (integer), Title (string), Duration (in seconds). Write a C program to perform the following operations: Insert songs into the playlist in the order they are added. Remove all songs whose duration is less than a user-specified value (Use head pointer for performing this operation). Display the updated playlist. Include a main() function to demonstrate all functionalities. (5)
- 10) A mobile app named "MIT-BLR Connect" maintains contact details of alumni in three different modules: i) Add Recent Contacts Module: Displays recently interacted alumni in order of contact - new ones added at the beginning. ii) Traverse Module: Allows viewing alumni details in forward direction. iii) Develop a module to print the data in reverse order. Develop C functions considering singly linked list as the suitable data structure (5)
- 11) Write a function to convert an infix expression to postfix. Include the following functions as per the prototype. i) int InStackPrecedence(char op) ; to return the precedence of operator within the stack ii) int IncomingPrecedence(char op); to return the precedence of the incoming operator. iii) void Infix_Postfix(char infix[], char postfix[]); to convert given infix expression to postfix. Main function not required. Note: The infix expression may include ONLY the following operators (listed in the order of their precedence): ; ^ (exponentiation) → right associative ; * (multiplication), / (division) → left associative ; + (addition), - (subtraction) → left associative (5)
- 12) Write a C function bool isDuplicate(char s[]) that detects duplicate parentheses in a given expression using a stack. (2)
- 13) Using the following nested calls, justify why a stack is preferred over a queue in handling function calls. Explain the role of activation records (stack frames).
void fun1() { printf("Inside fun1\n"); }
void fun2() { fun1(); printf("Inside fun2\n"); }
int main() { fun2(); printf("Inside main\n"); } (2)

End of question paper

Worked Solutions

Data Structures

Set A: Ensemble | CSE Semester 3

Companion to `ds-ensemble.pdf`. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Undo data structure

Correct option: (B). A stack removes the most recently pushed action first (LIFO).

2. First undo

Correct option: (D). C is the most recent edit and is at the top of the stack.

3. Redo stack order

Correct option: (A). The first undo pushes C onto redo; the second pushes B above C.

4. Redo operation

Correct option: (C). Redo pops the top action, B, and puts it back on undo.

5. Edit after undo

Correct option: (B). A new edit starts a new editing branch, so the previously undone future actions are discarded.

6. Nested employee structure

The outer structure stores employee details and the nested `Date` stores joining date. This program assumes a valid calendar date and a one-word name of at most 39 characters. The input-conversion count is checked.

```
#include <stdio.h>
typedef struct { int day, month, year; } Date;
typedef struct {
    int id;
    char name[40];
    Date joined;
} Employee;
int main(void) {
    Employee e;
    /* Input: id, one-word name, day, month, year. */
    if (scanf("%d %39s %d %d %d", &e.id, e.name,
            &e.joined.day, &e.joined.month,
            &e.joined.year) != 5) return 1;
    printf("%d %s %02d/%02d/%04d\n", e.id, e.name,
            e.joined.day, e.joined.month, e.joined.year);
    return 0;
}
```

7. Dynamic matrix and pointer arithmetic

A pointer to an array of three doubles provides contiguous dynamic storage for the three subjects. The expression $*(*(marks + i) + j)$ accesses one mark using pointer arithmetic. Time is $O(n)$ for three subjects and storage is $O(n)$.

```
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
int main(void) {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 1;
    if ((size_t)n > SIZE_MAX / (3 * sizeof(double))) return 1;
    double (*marks)[3] = malloc((size_t)n * sizeof *marks);
    if (!marks) return 1;
    for (int i = 0; i < n; ++i) {
        double sum = 0;
        for (int j = 0; j < 3; ++j) {
            if (scanf("%lf", *(marks + i) + j) != 1) {
                free(marks); return 1;
            }
            sum += (*(marks + i) + j);
        }
        printf("Student %d average: %.2f\n", i + 1, sum / 3);
    }
    free(marks);
    return 0;
}
```

8. Dynamic string array and concatenation

Allocate two fixed-capacity rows dynamically, then allocate the exact combined length plus one byte for the null terminator. The pointer `dst` advances through the result as each character is copied. Inputs must each fit in 99 characters before their newline.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main(void) {
    /* Two rows of up to 99 characters, including spaces. */
    char (*s)[100] = malloc(2 * sizeof *s);
    if (!s) return 1;
    if (!fgets(*s, 100, stdin) || !fgets(*(s + 1), 100, stdin)) {
        free(s); return 1;
    }
    *(*s + strcspn(*s, "\n")) = '\0';
    *(*s + 1) + strcspn(*(s + 1), "\n")) = '\0';
    char *result = malloc(strlen(*s) + strlen(*(s + 1)) + 1);
    if (!result) { free(s); return 1; }
    char *dst = result;
    for (int row = 0; row < 2; ++row)
        for (const char *src = *(s + row); *src; ++src)
            *dst++ = *src;
    *dst = '\0';
    puts(result);
    free(result); free(s);
    return 0;
}
```

9. Playlist insertion and deletion

Append preserves insertion order. A pointer to the current link permits deletion of the head and consecutive short songs without a separate head special case. Deletion and display take $O(n)$ time; this simple append scans to the tail and takes $O(n)$ per insertion. Enter 180 in the demonstration to retain only Evening.

```
#include <stdio.h>
#include <stdlib.h>
typedef struct Song {
    int id, duration;
    char title[80];
    struct Song *next;
} Song;

int append(Song **head, int id, const char *title, int duration) {
    Song *node = malloc(sizeof *node);
    if (!node) return 0;
    node->id = id; node->duration = duration; node->next = NULL;
    snprintf(node->title, sizeof node->title, "%s", title);
    Song **link = head;
    while (*link) link = &(*link)->next;
    *link = node;
    return 1;
}

void removeShort(Song **head, int limit) {
    Song **link = head;
    while (*link) {
        if ((*link)->duration < limit) {
            Song *old = *link;
            *link = old->next;
            free(old);
        } else link = &(*link)->next;
    }
}

void display(const Song *p) {
    for (; p; p = p->next)
        printf("%d %s %d\n", p->id, p->title, p->duration);
}

void clear(Song **head) {
    while (*head) {
        Song *old = *head; *head = old->next; free(old);
    }
}

int main(void) {
    Song *head = NULL;
    if (!append(&head, 1, "Dawn", 120) ||
        !append(&head, 2, "Evening", 240) ||
        !append(&head, 3, "Night", 150)) {
        clear(&head); return 1;
    }
    int limit;
    if (scanf("%d", &limit) != 1) { clear(&head); return 1; }
    removeShort(&head, limit);
    display(head);
    clear(&head);
    return 0;
}
```

10. Recent contacts and traversal

Insert at the head in $O(1)$ time. Forward and reverse printing take $O(n)$ time. The recursive reverse printer uses $O(n)$ call-stack space and does not change the list. Each call adds a contact record; moving an already present contact would additionally require locating and unlinking its previous node.

```
#include <stdio.h>
#include <stdlib.h>
typedef struct Contact {
    int id;
    char name[40];
    struct Contact *next;
} Contact;
int addRecent(Contact **head, int id, const char *name) {
    Contact *p = malloc(sizeof *p);
    if (!p) return 0;
    p->id = id;
    snprintf(p->name, sizeof p->name, "%s", name);
    p->next = *head; *head = p;
    return 1;
}
void forward(const Contact *p) {
    for (; p; p = p->next) printf("%d %s\n", p->id, p->name);
}
void reversePrint(const Contact *p) {
    if (!p) return;
    reversePrint(p->next);
    printf("%d %s\n", p->id, p->name);
}
```

11. Infix to postfix with precedence functions

Pop while in-stack precedence is at least incoming precedence. Giving incoming exponentiation precedence 6 and in-stack exponentiation precedence 5 keeps equal exponentiation operators on the stack, producing right associativity. Equal-precedence addition, subtraction, multiplication and division pop, producing left associativity. This C99 implementation handles valid expressions with single-character operands and optional parentheses.

```
#include <ctype.h>
#include <stddef.h>
#include <string.h>
int InStackPrecedence(char op) {
    switch (op) {
        case '+': case '-': return 1;
        case '*': case '/': return 3;
        case '^': return 5;
        case '(': return 0;
        default: return -1;
    }
}
int IncomingPrecedence(char op) {
    switch (op) {
        case '+': case '-': return 1;
        case '*': case '/': return 3;
        case '^': return 6; /* Do not pop an existing '^'. */
        case '(': return 7;
        default: return -1;
    }
}
/* Valid expressions, single-character operands; output has
   space for strlen(infix)+1 bytes. Parentheses are supported. */
void Infix_Postfix(char infix[], char postfix[]) {
    char stack[strlen(infix) + 1];
    size_t top = 0, out = 0;
    for (size_t i = 0; infix[i]; ++i) {
        char c = infix[i];
        if (isspace((unsigned char)c)) continue;
        if (isalnum((unsigned char)c)) postfix[out++] = c;
        else if (c == '(') stack[top++] = c;
        else if (c == ')') {
            while (top && stack[top - 1] != '(')
                postfix[out++] = stack[--top];
            if (top) --top;
        } else {
            while (top && stack[top - 1] != '(' &&
                InStackPrecedence(stack[top - 1]) >=
                IncomingPrecedence(c))
                postfix[out++] = stack[--top];
            stack[top++] = c;
        }
    }
    while (top) postfix[out++] = stack[--top];
    postfix[out] = '\0';
}
```

For A^B^C , the output is $ABC^{\wedge\wedge}$. For $A-B-C$, it is $AB-C-$. Each token is pushed and popped at most once, so time and auxiliary space are $O(n)$.

12. Duplicate parentheses

Here duplicate means a redundant pair enclosing no operator at its own nesting level, including an extra wrapper around a parenthesized expression. The input is assumed syntactically valid and balanced. Pop a closed group, check whether an operator occurred, and reduce a useful group to one operand marker. Time and stack space are $O(n)$.

```
#include <stdbool.h>
#include <ctype.h>
#include <string.h>
/* Input is a syntactically valid, balanced expression.
   Redundant pair means a pair enclosing no operator at its
   own nesting level: (a), ((a+b)), and () are detected. */
bool isDuplicate(char s[]) {
    char stack[strlen(s) + 1];
    size_t top = 0;
    for (size_t i = 0; s[i]; ++i) {
        char c = s[i];
        if (isspace((unsigned char)c)) continue;
        if (c != ')') stack[top++] = c;
        else {
            bool hasOperator = false;
            while (top && stack[top - 1] != '(') {
                char v = stack[--top];
                if (strchr("+-*/^", v)) hasOperator = true;
            }
            if (top) --top;
            if (!hasOperator) return true;
            stack[top++] = 'x'; /* One reduced subexpression. */
        }
    }
    return false;
}
```

Examples: ((a+b)) returns true; (a+b) returns false; (a) returns true under the stated convention.

13. Activation records and LIFO

Calls push frames in this order: main, fun2, fun1. Returns remove frames in the reverse order: fun1, fun2, main. The output is:

```
Inside fun1
Inside fun2
Inside main
```

A frame holds the return address, parameters, local variables and saved execution state as required by the calling convention. A queue would remove the oldest caller first, although that caller is suspended awaiting its callee, so it does not represent normal nested call/return order.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A 900-byte payload passes through five layers, each adding a 20-byte header. With no trailers, what is the transmitted size? (1)
(A) 920 bytes
(B) 1000 bytes
(C) 1020 bytes
(D) 1100 bytes
- 2) A transmitted unit contains 900 payload bytes and 100 header bytes. What percentage of the total is headers? (1)
(A) 9%
(B) 11.11%
(C) 90%
(D) 10%
- 3) What is the polynomial degree of CRC generator 10011? (1)
(A) 3
(B) 5
(C) 4
(D) 2
- 4) How many zeros are appended to the data before CRC division using generator 10011? (1)
(A) 4
(B) 5
(C) 3
(D) 1
- 5) Which operation performs subtraction in CRC modulo-2 division? (1)
(A) AND
(B) XOR
(C) OR
(D) Decimal subtraction

Section B: Theory questions (25 marks)

- 6) Compare Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) in terms of how each modifies the carrier signal, their bandwidth efficiency, and susceptibility to noise. Analyse which technique is most suitable for transmission over a noisy telephone channel and justify your choice. (6)

- 7) Define data rate and bandwidth. A square wave is approximated by (6)

$$s(t) = \frac{4}{\pi} \left[\sin(2\pi \cdot 2 \cdot 10^6 t) + \frac{1}{3} \sin(2\pi \cdot 6 \cdot 10^6 t) + \frac{1}{5} \sin(2\pi \cdot 10 \cdot 10^6 t) \right].$$

Assume two bits per fundamental cycle and bandwidth equal to highest minus lowest retained frequency. Calculate the data rate and bandwidth. Triple all frequencies and find the new values. Explain how bandwidth affects data rate.

- 8) For the given binary data 101000101010, draw the following line coding scheme. (Assume previous pulse was negative) i. Differential Manchester ii. Bipolar AMI For Differential Manchester, use a boundary transition for 0 and none for 1; initial level is negative. A mid-bit transition always occurs. (2)
- 9) For the given binary data 111000101010, draw the following line coding scheme. (Assume previous pulse was negative) i) Manchester ii) Bipolar AMI Manchester convention: 1 is low-to-high and 0 is high-to-low at mid-bit. (2)
- 10) A company network consists of three departments connected as follows: Sales and HR are connected through a hub, HR and IT are connected through a switch, and the switch is connected to a router for internet access. An employee in Sales sends a file to an employee in IT. Analyse how the hub, switch, and router handle this transmission. (2)
- 11) A sender uses a window of 3 to transmit frames 0, 1 and 2. Frame 1 is lost, while 0 and 2 arrive. ACKs are immediate and never lost; no new frames are available. In go-back-N, discard out-of-order frames and retransmit all outstanding frames at timeout. In selective repeat, buffer out-of-order frames and retransmit only the unacknowledged frame. For each protocol, list retransmissions and calculate useful frames divided by total data transmissions. (3)
- 12) Calculate the total time required to transmit 1,000,000 bits using the Stop-and-Wait ARQ protocol when each frame carries 2000 bits, the sender and receiver are 5000 km apart, and the propagation speed is 2×10^8 m/s. Ignore transmission, waiting, and processing delays, and assume no frames are lost or damaged Count one full round-trip per frame, including the final ACK. (2)
- 13) An Ethernet segment has transmission speed 10^8 bits/second and maximum length 1500 metres. If propagation speed is 3×10^8 metres/second, calculate the minimum frame size in bits required for collision detection. (2)

End of question paper

Worked Solutions

Data Communication and Computer Networks

Set A: Ensemble | CSE Semester 3

Companion to dccn-ensemble.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Layered headers

Correct option: (B). The total is $900 + 5(20) = 1000$ bytes.

2. Header percentage

Correct option: (D). The fraction is $100/(900 + 100) = 0.10 = 10\%$.

3. CRC degree

Correct option: (C). The leading term is x^4 , so the generator has degree 4.

4. CRC appended zeros

Correct option: (A). Append as many zeros as the generator degree, which is 4.

5. Modulo-2 subtraction

Correct option: (B). Modulo-2 subtraction is XOR, with no carry or borrow.

6. ASK, FSK and PSK

ASK changes carrier amplitude, FSK changes carrier frequency, and PSK changes carrier phase. Binary FSK commonly uses two separated frequencies and needs more bandwidth than binary PSK under comparable simple pulse shaping. ASK and BPSK can have comparable nominal bandwidth for the same bit rate and shaping; exact occupied bandwidth depends on the implementation. ASK is directly vulnerable to amplitude disturbances. Constant-envelope FSK and PSK are less affected by such disturbances, although phase/frequency noise and receiver synchronization still matter.

For a coherent receiver in an approximately additive-noise, bandwidth-limited telephone channel, BPSK is a defensible choice: it uses antipodal signals, offers good bit-error performance at a given energy per bit, and avoids the tone separation of binary FSK. A noncoherent, very simple receiver could instead favor FSK. The noise model and implementation must be stated rather than claiming one scheme is always best.

7. Data rate and bandwidth

Data rate is information bits transmitted per second. Here bandwidth means the span from the lowest to highest retained frequency. The three frequencies are 2, 6 and 10 MHz.

$$R_b = 2f_0 = 2(2 \text{ MHz}) = \boxed{4 \text{ Mbps}}, \quad B = 10 - 2 = \boxed{8 \text{ MHz}}.$$

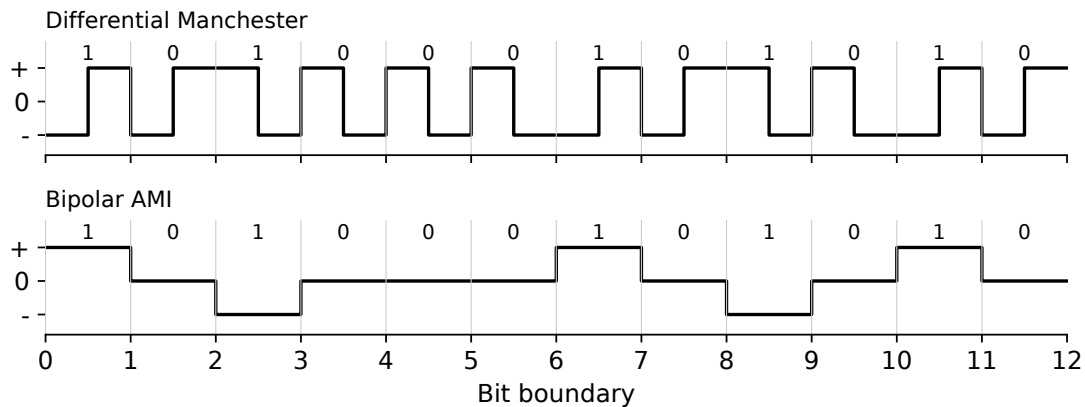
After tripling all frequencies, the components are 6, 18 and 30 MHz:

$$R'_b = \boxed{12 \text{ Mbps}}, \quad B' = \boxed{24 \text{ MHz}}.$$

For this fixed waveform model, both increase by a factor of three. A greater usable channel bandwidth permits faster signal changes, but achievable reliable data rate also depends on noise, signal levels and coding. The two-bits-per-cycle assumption is essential; it is not inferred from a sinusoid alone.

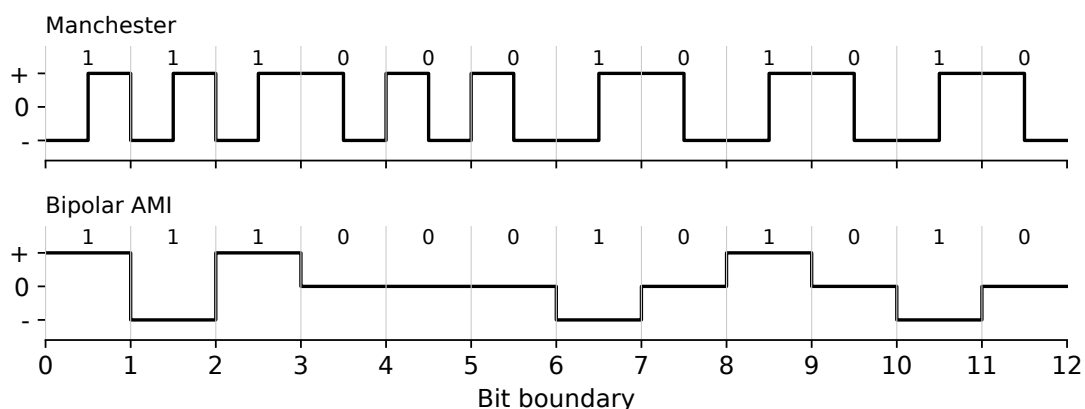
8. Differential Manchester and AMI

Start at a negative level. For differential Manchester, a 0 causes a transition at the start of its bit, a 1 does not, and every bit has a mid-bit transition. For AMI, zeros have zero level and successive ones alternate positive and negative, starting positive because the prior nonzero pulse was negative. The diagram labels each bit boundary.



9. Manchester and AMI

Use low-to-high for Manchester 1 and high-to-low for Manchester 0. AMI ones alternate polarity, with the first one positive; zeros are at zero level.



10. Hub, switch and router

The hub repeats the Sales signal out its other ports without inspecting a destination address. HR and the attached switch therefore see it. The switch learns the source MAC address on the ingress port and forwards the frame only to the known IT destination port. If that MAC is unknown, it floods within the VLAN except back to the ingress port. A destination learned on the ingress port is filtered.

If Sales and IT are in the same IP subnet and VLAN, their local traffic need not pass through the Internet router. If they are in different subnets/VLANs, the sender uses its default gateway and the router must route between those networks. A router forwards according to destination IP and installs a new link-layer header on the outgoing link. Merely being connected to the Internet does not make it part of every local transfer.

11. ARQ retransmission counts

Initially frames 0, 1 and 2 are transmitted, with frame 1 lost.

Protocol	Receiver handling of 2	Retransmitted	Total sends	Useful/total
Go-back-N	Discard	1, 2	5	3/5 = 0.60
Selective repeat	Buffer	1	4	3/4 = 0.75

Go-back-N has already accepted frame 0; on timeout it resends the outstanding suffix 1 and 2. Selective repeat acknowledges and buffers frame 2, then needs only frame 1 before delivering 1 and 2 in order. Counts use the clarified three-frame scenario printed in this paper, not the older ambiguous repeated-loss wording.

12. Propagation-limited stop-and-wait time

Number of frames = $10^6/2000 = 500$.

$$t_p = \frac{5000 \times 10^3}{2 \times 10^8} = 0.025 \text{ s}, \quad RTT = 0.05 \text{ s}.$$

Counting one complete round trip for each frame, including the final ACK, gives

$$T = 500(0.05) = \boxed{25 \text{ s}}.$$

Transmission, ACK serialization and processing times are neglected as requested.

13. Collision-detection frame length

$$t_p = \frac{1500}{3 \times 10^8} = 5 \mu\text{s}, \quad L_{\min} = R(2t_p) = 10^8(10 \times 10^{-6}) = \boxed{1000 \text{ bits}}.$$

This is 125 bytes under the question's propagation-only model. The sender must still be transmitting when a far-end collision can propagate back.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) What is the range of 5-bit two's-complement integers? (1)
(A) -15 to 15
(B) -16 to 15
(C) -16 to 16
(D) 0 to 31
- 2) What is the 5-bit two's-complement representation of -9 ? (1)
(A) 01001
(B) 11001
(C) 10110
(D) 10111
- 3) Which addition overflows the 5-bit two's-complement range? (1)
(A) $-9 + 5$
(B) $6 - 13$
(C) $11 + 7$
(D) $-4 + 3$
- 4) Which arrays are programmable in a programmable logic array (PLA)? (1)
(A) Both AND and OR
(B) Only AND
(C) Only OR
(D) Neither
- 5) A PLA implements $F_1 = AB + AC$ and $F_2 = AB + BC$. Which product term can be shared? (1)
(A) AC
(B) AB
(C) BC
(D) ABC

Section B: Theory questions (25 marks)

- 6) Simplify $F(A, B, C, D) = \Pi M(0, 3, 5, 6, 8, 10)$, with don't-care terms $d(2, 13, 14)$, and implement it using NOR gates only. Take A as the most significant bit. (3)
- 7) Design a 3-bit magnitude comparator for unsigned binary numbers $A = a_2a_1a_0$ and $B = b_2b_1b_0$. Derive the Boolean expressions for $A > B$, $A = B$ and $A < B$, and draw the equivalent logic circuit. (3)
- 8) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)
- 9) Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If $E = 0$, the circuit remains in the same state regardless of the value of F. When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (5)
- 10) Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. (3)
- 11) Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same. (3)
- 12) Write behavioral Verilog code that represents an eight-bit ring counter. (2)
- 13) Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing. (3)

End of question paper

Worked Solutions

Digital Systems and Computer Organization

Set A: Ensemble | CSE Semester 3

Companion to `dsc0-ensemble.pdf`. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Signed range

Correct option: (B). For n bits, the range is -2^{n-1} to $2^{n-1} - 1$, giving -16 to 15 .

2. Negative representation

Correct option: (D). Invert `01001` to get `10110`, then add 1 to obtain `10111`.

3. Signed overflow

Correct option: (C). $11 + 7 = 18$ exceeds the maximum signed value 15 ; the other results lie in range.

4. PLA arrays

Correct option: (A). A PLA has programmable AND and OR arrays for constructing and combining product terms.

5. Shared product term

Correct option: (B). AB appears in both outputs and can feed both OR connections.

6. POS minimization and NOR implementation

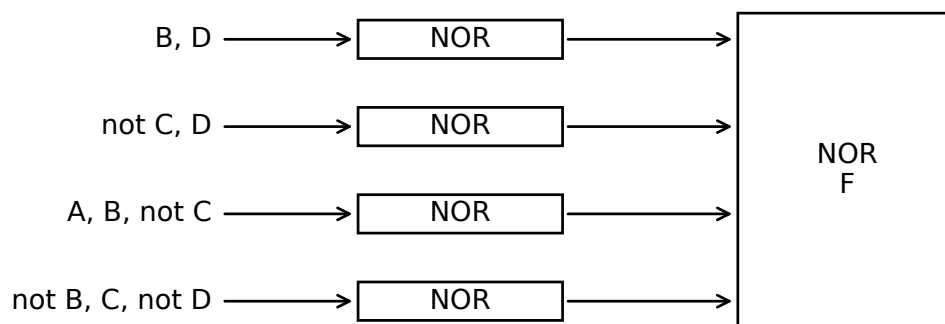
Use a K-map with rows AB and columns CD in Gray order 00, 01, 11, 10. X denotes a don't-care.

AB / CD	00	01	11	10
00	0	1	0	X
01	1	0	1	0
11	1	X	1	X
10	0	1	1	0

Group zeros with useful don't-cares: $\{0, 2, 8, 10\}$, $\{2, 6, 10, 14\}$, $\{2, 3\}$ and $\{5, 13\}$. These give

$$F = (B + D)(\bar{C} + D)(A + B + \bar{C})(\bar{B} + C + \bar{D}).$$

Each first-level NOR gives the complement of one sum. A final NOR of those four outputs produces their uncomplemented product by De Morgan's law. Generate $\bar{B}$, $\bar{C}$, $\bar{D}$ with tied-input NOR gates: $\bar{X} = \text{NOR}(X, X)$.



7. Three-bit magnitude comparator

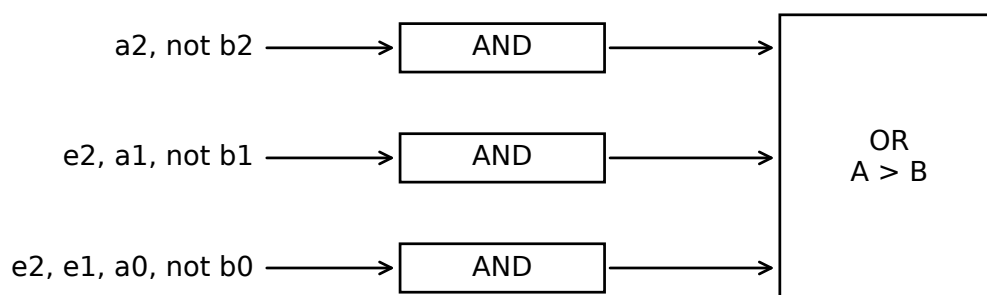
Define $e_i = a_i \text{ XNOR } b_i$. The most significant unequal bit decides the ordering:

$$G = a_2\bar{b}_2 + e_2a_1\bar{b}_1 + e_2e_1a_0\bar{b}_0,$$

$$E_q = e_2e_1e_0,$$

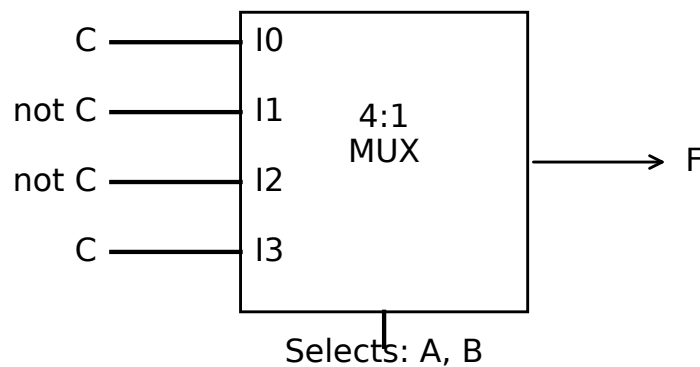
$$L = \bar{a}_2b_2 + e_2\bar{a}_1b_1 + e_2e_1\bar{a}_0b_0.$$

Generate each e_i with an XNOR gate and each complemented input with NOT. The following AND-OR network implements G. Implement L with the same network after exchanging a and b. A three-input AND of the XNOR outputs implements E_q . These three networks together form the comparator.



8. Three-input XOR using a multiplexer

With A and B as select bits (A most significant), $AB = 00, 01, 10, 11$ produces $C, \bar{C}, \bar{C}, C$, respectively. Thus $I_0 = C, I_1 = \bar{C}, I_2 = \bar{C}, I_3 = C$. One inverter supplies both complemented-C inputs.



9. Enabled synchronous up/down counter

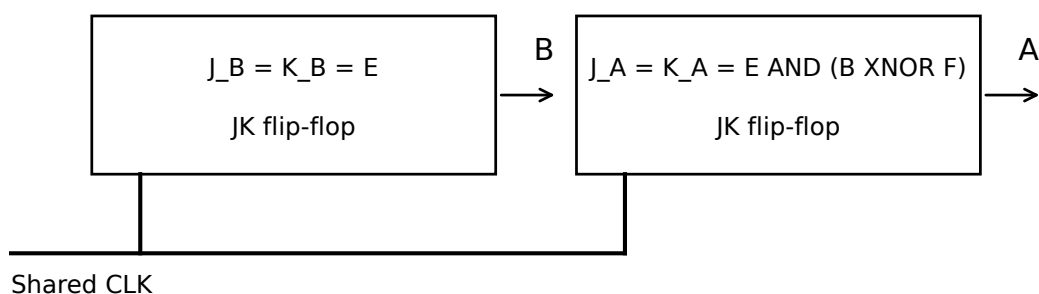
Let A be the most significant state bit and B the least significant. The state table is:

Present AB	E=0, either F	E=1, F=1	E=1, F=0
00	00	01	11
01	01	10	00
10	10	11	01
11	11	00	10

A JK flip-flop toggles when $J = K = 1$ and holds when $J = K = 0$. B toggles whenever enabled. A toggles on an upward carry ($B=1, F=1$) or downward borrow ($B=0, F=0$).

$$J_B = K_B = E, \quad J_A = K_A = E(BF + \bar{B}\bar{F}).$$

Equivalently, the second excitation is E AND (B XNOR F). Connect both flip-flops to the same active clock edge and use the following excitation network.

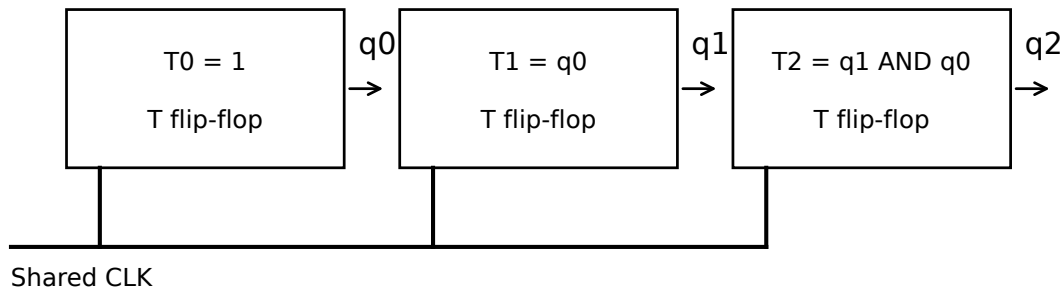


10. Three-bit synchronous T counter

For $q_2q_1q_0$, the least significant bit toggles every clock; bit 1 toggles when $q_0 = 1$; bit 2 toggles when $q_1q_0 = 11$.

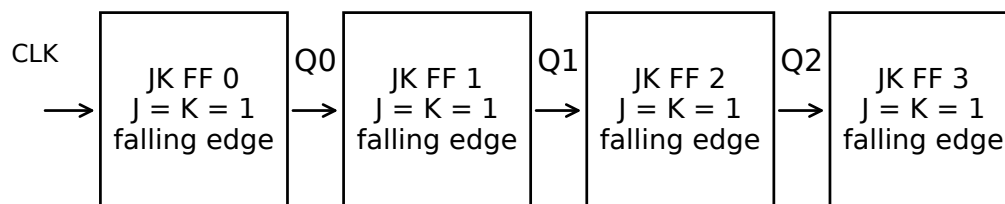
$$T_0 = 1, \quad T_1 = q_0, \quad T_2 = q_1q_0.$$

All three T flip-flops use the same clock. Feed q_1, q_0 through an AND gate to T_2 . The sequence is 000, 001, 010, 011, 100, 101, 110, 111, 000.



11. Four-bit asynchronous ripple counter

Use four falling-edge-triggered JK flip-flops with $J=K=1$. Apply the external clock to stage 0 and connect each stage's Q to the next stage's clock, as shown. Clear all stages to 0 before counting.



The output $q_3q_2q_1q_0$ counts 0000 through 1111 and repeats. Output frequencies are $f/2, f/4, f/8, f/16$. Since changes propagate stage by stage, transitions such as 0111 to 1000 pass through intermediate states. Decoding before the ripple settles can produce glitches. The falling-edge convention is necessary for this Q-to-clock up-counter connection.

12. Eight-bit ring counter

The question leaves reset details open; choose an active-high synchronous reset to load a single 1. This makes the initial state deterministic. Each clock rotates left, including the wrap from bit 7 to bit 0.

```
module ring8(input wire clk, input wire reset,
             output reg [7:0] q);
    always @(posedge clk) begin
        if (reset) q <= 8'b00000001;
        else q <= {q[6:0], q[7]};
    end
endmodule
```

The cycle is 00000001, 00000010, 00000100, 00001000, 00010000, 00100000, 01000000, 10000000, then 00000001. Reset must be asserted to enter this one-hot cycle; an all-zero state would otherwise remain zero.

13. Byte and word addressing

A memory address identifies an addressable unit of storage. An ISA defines how effective addresses are formed and how load/store instructions interpret them. In byte addressing, consecutive addresses select consecutive bytes. A 32-bit word therefore spans four byte addresses, and consecutive aligned words differ by 4. In word addressing, consecutive addresses select entire words, so consecutive words differ by 1. For 2^a addressable units, capacity is 2^a bytes in the byte-addressed case and $2^a w$ bytes for w-byte word addressing. Word size and address width are separate concepts.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

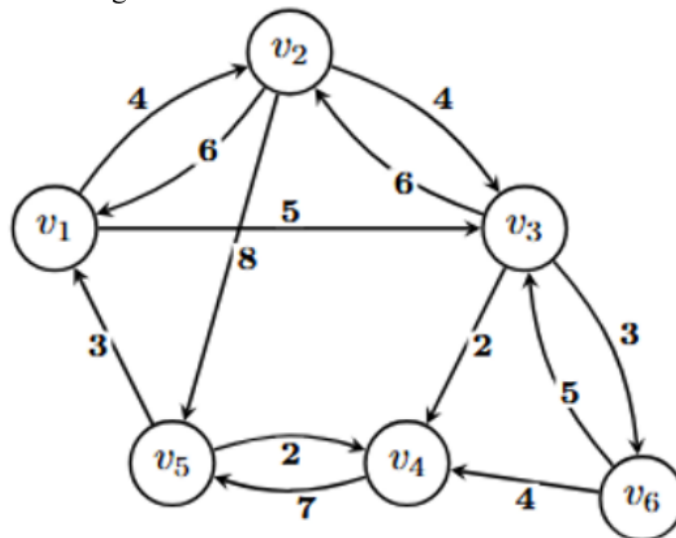
Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which pair of conditions defines a complement b of a in a Boolean algebra? (1)
(A) $a \vee b = 0, a \wedge b = 1$
(B) $a \vee b = 1, a \wedge b = 0$
(C) $a \vee b = a, a \wedge b = b$
(D) $a = b$
- 2) How many complements does each element of a Boolean algebra have? (1)
(A) None
(B) At least two
(C) Infinitely many
(D) Exactly one
- 3) Which expression equals $a \vee (\neg a \wedge b)$? (1)
(A) $a \vee b$
(B) $a \wedge b$
(C) $\neg a \vee b$
(D) a
- 4) Which expression is in disjunctive normal form (a disjunction of conjunctions of literals)? (1)
(A) $(x \vee y) \wedge z$
(B) $\neg(x \vee y)$
(C) $(x \wedge y) \vee (\neg x \wedge z)$
(D) $(x \vee y) \wedge (\neg x \vee z)$
- 5) Which expression is in conjunctive normal form (a conjunction of disjunctions of literals)? (1)
(A) $(x \wedge y) \vee z$
(B) $(x \vee y) \wedge (\neg x \vee z)$
(C) $\neg(x \wedge y)$
(D) $(x \wedge y) \vee (\neg x \wedge z)$

Section B: Theory questions (25 marks)

- 6) Draw the Hasse diagram for $A = \{1, 2, 3, 12, 18, 36\}$, ordered by divisibility: a is related to b iff a divides b . Is it a lattice? Justify. List all antichains of length more than one. (5)
- 7) How many ways are there to select a total of 15 items from three types, subject to these limits? (4)
Type A: 1 to 4 items; Type B: 3 to 7 items; Type C: 5 to 10 items.
- 8) Show that the number of partitions of n with no part smaller than 3 equals the number of partitions of $n+3$ whose three largest parts are consecutive integers. (3)
- 9) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)
- 10) Show that if G is a regular self-complementary graph of order n , then $n = 4t + 1$ for some integer t . (3)
- 11) If G has p vertices and minimum degree $\delta(G) \geq (p - 1)/2$, prove that G is connected. (3)
- 12) Use Dijkstra's algorithm to find the shortest paths from v_1 to every other vertex in the directed network below. Show the steps. (4)



End of question paper

Worked Solutions

Discrete Mathematical Structures

Set A: Ensemble | CSE Semester 3

Companion to dms-ensemble.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Complement conditions

Correct option: (B). A complement joins with the element to give 1 and meets with it to give 0.

2. Complement uniqueness

Correct option: (D). Complement existence is a Boolean-algebra property, and distributivity gives uniqueness.

3. Boolean simplification

Correct option: (A). Distribution gives $(a \vee \neg a) \wedge (a \vee b) = 1 \wedge (a \vee b) = a \vee b$.

4. Disjunctive normal form

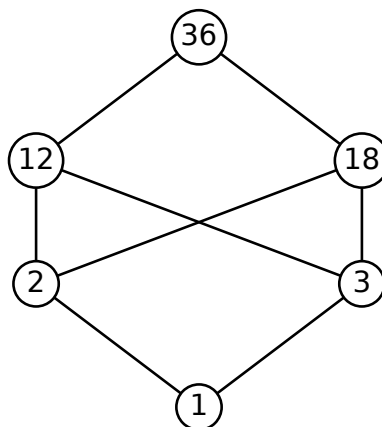
Correct option: (C). $(x \wedge y) \vee (\neg x \wedge z)$ is an OR of two AND terms of literals.

5. Conjunctive normal form

Correct option: (B). $(x \vee y) \wedge (\neg x \vee z)$ is an AND of two OR clauses of literals.

6. Divisibility poset and antichains

The cover relations are $(1, 2), (1, 3), (2, 12), (2, 18), (3, 12), (3, 18), (12, 36), (18, 36)$. They determine the Hasse diagram below; line crossings without a labelled vertex are not additional elements.



This poset is **not a lattice**. The common upper bounds of 2 and 3 are 12, 18 and 36. The two minimal common upper bounds 12 and 18 are incomparable, so there is no least upper bound. Equivalently, 12 and 18 have no greatest lower bound in the set: 2 and 3 are incomparable maximal common lower bounds. It is not valid to import gcd 6 into the set, since 6 is absent.

The complete list of antichains with more than one element is $\{2, 3\}, \{12, 18\}$. Every other distinct pair is comparable, so there are no larger antichains.

7. Bounded item selection

Let $u = A - 1$, $v = B - 3$, $w = C - 5$. Then $u + v + w = 6$, with $0 \leq u \leq 3$, $0 \leq v \leq 4$, $0 \leq w \leq 5$. The unrestricted nonnegative count is $\binom{8}{2} = 28$.

Subtract violations: $u \geq 4$ contributes $\binom{4}{2} = 6$, $v \geq 5$ contributes $\binom{3}{2} = 3$, and $w \geq 6$ contributes $\binom{2}{2} = 1$. Intersections are impossible because their minimum sums exceed 6. Thus

$$28 - 6 - 3 - 1 = 18.$$

Items of the same type are indistinguishable, so this counts feasible triples of quantities, not arrangements.

8. Partition bijection

For positive n , let λ be a partition of n with every part at least 3. In its Ferrers diagram, the first three columns have the same height k . Its conjugate partition $\mu = \lambda'$ therefore satisfies $\mu_1 = \mu_2 = \mu_3 = k$. Define a new partition

$$\nu = (\mu_1 + 2, \mu_2 + 1, \mu_3, \mu_4, \dots).$$

Its total is $n + 3$, and its three largest parts are $k + 2, k + 1, k$, which are consecutive.

Conversely, if a partition of $n + 3$ has largest parts $k + 2, k + 1, k$, subtract 2 from the first part and 1 from the second. The resulting partition has three equal largest parts k and total n . Conjugating it gives a partition of n in which every row has at least three cells, hence every part is at least 3. These constructions undo each other, proving the claimed equality. For $n=1$ or 2 , both sets are empty.

9. The 75th Fike permutation

Use the original descending-exchange Fike convention, with the original arrangement 01234 counted as permutation 1. Enumerate (d_2, d_3, d_4, d_5) with $d_k = k, k - 1, \dots, 1$, and d_5 varying fastest. For each tuple, start from 01234 and exchange positions k and d_k in increasing k order, using one-based positions.

The zero-based index is $75 - 1 = 74$. The block weights for these four choices are 60, 20, 5 and 1:

$$74 = 1(60) + 0(20) + 2(5) + 4(1).$$

Thus $(d_2, d_3, d_4, d_5) = (2 - 1, 3 - 0, 4 - 2, 5 - 4) = (1, 3, 2, 1)$.

Exchange	Arrangement
Start	01234
Positions 2 and 1	10234
Positions 3 and 3	10234
Positions 4 and 2	13204
Positions 5 and 1	43201

Hence the answer under this convention is **43201**.

Convention reference: J. S. Rohl, "Programming Improvements to Fike's Algorithm for Generating Permutations," The Computer Journal 19(2), 156–159 (1976), Figure 1 and recursive procedure. <https://doi.org/10.1093/comjnl/19.2.156>. The convention is stated explicitly because Fike order is not lexicographic order.

10. Regular self-complementary graph

Assume a nonempty simple graph. If G is r -regular on n vertices, its complement is $(n - 1 - r)$ -regular. Isomorphism to its complement implies

$$r = n - 1 - r, \quad r = (n - 1)/2.$$

Therefore n is odd. The handshake lemma says $nr = 2|E|$ is even. Since n is odd, r must be even; write $r = 2t$. Substitution gives $n = 2r + 1 = \boxed{4t + 1}$, with $t \geq 0$.

11. Minimum degree forces connectivity

Suppose G is disconnected, and a component has s vertices. A vertex in that component has degree at most $s - 1$, so

$$s - 1 \geq \delta(G) \geq (p - 1)/2, \quad s \geq (p + 1)/2.$$

Every component must satisfy this bound. Two components would already contain at least $p+1$ vertices, contradicting the total p . Hence G is connected. This argument assumes the usual simple-graph setting; a one-vertex graph is connected directly.

12. Dijkstra on the directed source network

Respect arrow direction. The outgoing edges are: $v_1 \rightarrow v_2 : 4$, $v_1 \rightarrow v_3 : 5$; $v_2 \rightarrow v_1 : 6$, $v_2 \rightarrow v_3 : 4$, $v_2 \rightarrow v_5 : 8$; $v_3 \rightarrow v_2 : 6$, $v_3 \rightarrow v_4 : 2$, $v_3 \rightarrow v_6 : 3$; $v_4 \rightarrow v_5 : 7$; $v_5 \rightarrow v_1 : 3$, $v_5 \rightarrow v_4 : 2$; $v_6 \rightarrow v_3 : 5$, $v_6 \rightarrow v_4 : 4$. The table records tentative distances after relaxing outgoing edges of each settled vertex.

Settled	d_1	d_2	d_3	d_4	d_5	d_6
Initial	0	∞	∞	∞	∞	∞
v_1	0	4	5	∞	∞	∞
v_2	0	4	5	∞	12	∞
v_3	0	4	5	7	12	8
v_4	0	4	5	7	12	8
v_6	0	4	5	7	12	8
v_5	0	4	5	7	12	8

Destination	Shortest path	Length
v_1	v_1	0
v_2	$v_1 \rightarrow v_2$	4
v_3	$v_1 \rightarrow v_3$	5
v_4	$v_1 \rightarrow v_3 \rightarrow v_4$	7
v_5	$v_1 \rightarrow v_2 \rightarrow v_5$	12
v_6	$v_1 \rightarrow v_3 \rightarrow v_6$	8

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) What is the mean of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 10
- 2) What is the median of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 9
- 3) The six observations 4, 6, 6, 8, 10, 14 have mean 8 and squared-deviation sum 64. What is their population variance? (1)
(A) $64/6$
(B) $64/5$
(C) $\sqrt{64/6}$
(D) 8
- 4) A distribution has mean 8, median 7 and positive standard deviation. What is the sign of Pearson's second skewness coefficient? (1)
(A) Zero
(B) Negative
(C) Undefined
(D) Positive
- 5) Which measure of central tendency is generally less affected by an extreme observation? (1)
(A) Arithmetic mean
(B) Median
(C) Midrange
(D) All equally affected

Section B: Theory questions (25 marks)

- 6) Find the five-number summary and interquartile range of 3, 5, 7, 8, 9, 11, 13, 16, 30. Exclude the median when forming the lower and upper halves. Use the 1.5 *IQR* rule to identify outliers and sketch a box plot. (3)
- 7) A table contains customer ID, age, monthly income and annual income. Give one suitable treatment for missing ages and explain how you would detect inconsistency between the two income fields. (2)
- 8) A filling machine is claimed to dispense a mean of 500 ml. A random sample of 36 bottles has mean 496 ml. The population standard deviation is known to be 12 ml. Perform a two-sided *z*-test at the 5% level. State the hypotheses, statistic, decision and conclusion. Use critical values ± 1.96 . (5)
- 9) Two judges rank six projects. Judge A assigns ranks (1, 2, 3, 4, 5, 6) and judge B assigns (2, 1, 4, 3, 6, 5). Calculate Spearman's rank correlation coefficient and interpret the result. (3)
- 10) Partition 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40 into three equal-frequency bins. Replace each observation by its bin mean and report the three resulting bin means. (2)
- 11) Test independence between training and passing an assessment using Pearson's chi-square test at 5%. Of 60 trained students, 48 pass and 12 fail; of 40 untrained students, 22 pass and 18 fail. Show expected frequencies and the statistic. Use $\chi^2_{0.05,1} = 3.841$; do not apply a continuity correction. (5)
- 12) Three independent normal populations have equal variances. Samples are A: 4, 5, 6, B: 7, 8, 9, C: 10, 11, 12. The between-group sum of squares is 54 and within-group sum of squares is 6. Complete the one-way ANOVA degrees of freedom, mean squares and *F* statistic. Use $F_{0.05;(2,6)} = 5.143$ and state your conclusion. (3)
- 13) Distinguish a Type I error from a Type II error in the context of the filling-machine test in Question 8. (2)

End of question paper

Worked Solutions

Data Analytics

Set B: Original | CSE Semester 3

Companion to da-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Arithmetic mean

Correct option: (C). The sum is 48 and $48/6 = 8$.

2. Median

Correct option: (B). The middle values are 6 and 8, so the median is $(6 + 8)/2 = 7$.

3. Population variance

Correct option: (A). Population variance divides the squared-deviation sum by N : $64/6$.

4. Pearson skewness sign

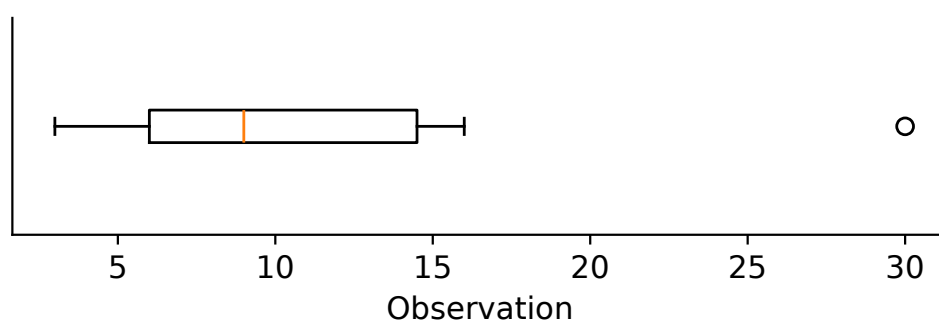
Correct option: (D). $3(\bar{x} - \text{median})/\sigma = 3(8 - 7)/\sigma > 0$.

5. Resistance to extremes

Correct option: (B). The median depends on the middle ranks and is generally more resistant to extremes.

6. Five-number summary and outlier

The median is 9. Excluding it gives lower half (3, 5, 7, 8) and upper half (11, 13, 16, 30). Thus $Q_1 = 6$, $Q_3 = 14.5$, and the five-number summary is (3, 6, 9, 14.5, 30). $IQR = 8.5$; the fences are -6.75 and 27.25 . Therefore 30 is an outlier. The modified box plot has whiskers at 3 and 16, with 30 plotted separately.



7. Missing and inconsistent data

Impute missing ages with the median of a relevant, sufficiently large customer group and retain a missingness flag. Estimate imputation values from training data only if building a predictive model. Flag rows where annual income differs materially from $12 \times$ monthly income after aligning currency, units, gross/net definitions and time period. Investigate such rows rather than automatically overwriting them.

8. Filling-machine mean test

$H_0 : \mu = 500$ ml, $H_1 : \mu \neq 500$ ml. The standard error is $12/\sqrt{36} = 2$ ml.

$$z = \frac{496 - 500}{2} = \boxed{-2.00}.$$

Since $|-2| > 1.96$, reject H_0 at 5%. There is evidence that the machine's mean fill differs from 500 ml; the sample suggests underfilling. The two-sided p -value is about 0.0455.

9. Rank correlation

The six rank differences are $(-1, 1, -1, 1, -1, 1)$, giving $\sum d_i^2 = 6$.

$$r_s = 1 - \frac{6(6)}{6(6^2 - 1)} = \boxed{\frac{29}{35} = 0.8286}.$$

There is strong positive rank agreement, although each adjacent pair is reversed.

10. Equal-frequency smoothing

Bin	Observations	Mean
1	10,12,15,18	$55/4 = 13.75$
2	20,22,25,28	$95/4 = 23.75$
3	30,32,35,40	$137/4 = 34.25$

The smoothed sequence is four copies of 13.75, followed by four copies of 23.75 and four copies of 34.25.

11. Training and passing

H_0 : training and passing are independent. Row totals are 60 and 40; column totals are 70 and 30. Expected counts are

$$E = \begin{pmatrix} 42 & 18 \\ 28 & 12 \end{pmatrix}.$$

$$\chi^2 = \frac{(48 - 42)^2}{42} + \frac{(12 - 18)^2}{18} + \frac{(22 - 28)^2}{28} + \frac{(18 - 12)^2}{12} = \boxed{7.1429}.$$

With one degree of freedom, $7.1429 > 3.841$, so reject independence. Training and passing are associated. The pass rates are 80% and 55%; causation requires an appropriate study design.

12. One-way ANOVA

Source	SS	df	MS	F
Between	54	2	27	27
Within	6	6	1	
Total	60	8		

Test $H_0 : \mu_A = \mu_B = \mu_C$. Since $F = 27 > 5.143$, reject H_0 at 5%. At least one population mean differs. Pairwise differences require further testing.

13. Type I and Type II errors

A Type I error is rejecting $\mu = 500$ when the machine's true mean is 500 ml. A Type II error is failing to reject $\mu = 500$ when the true mean differs from 500 ml. The 5% significance level controls the Type I error probability under the test assumptions; it does not specify the Type II probability.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which C keyword defines a structure type? (1)
(A) `class`
(B) `union`
(C) `struct`
(D) `enum`
- 2) Object `s` has a nested structure member `admitted` with an integer member `year`. How do you access that `year`? (1)
(A) `s.admitted.year`
(B) `s->admitted.year`
(C) `s.year.admitted`
(D) `s::admitted::year`
- 3) How many non-null characters can a null-terminated string in `char name[40]` hold? (1)
(A) 40
(B) 41
(C) 38
(D) 39
- 4) Which `scanf` format limits a single word to fit in `char name[40]`, including its null terminator? (1)
(A) `%40s`
(B) `%39s`
(C) `%s`
(D) `%41s`
- 5) If `p` points to a structure with integer member `roll`, which expression accesses that member? (1)
(A) `p.roll`
(B) `*p.roll`
(C) `p->roll`
(D) `p::roll`

Section B: Theory questions (25 marks)

- 6) Write a C function that dynamically allocates a contiguous $r \times c$ integer matrix and returns its base pointer, or NULL on failure. Reject nonpositive dimensions. Show how element (i, j) is accessed using pointer arithmetic. Assume the requested size fits in `size_t`. (3)
- 7) Explain a dangling pointer and a memory leak using one short C example of each. State how each error can be prevented. (2)
- 8) A singly linked list stores integer values. Define its node structure and write `void deleteAll(Node **head, int key)` to delete every node whose value equals `key`. Handle an empty list, consecutive matches and matches at the head. Release deleted nodes. (5)
- 9) Write a C function to reverse a singly linked list in place using three pointers. Return the new head and state its time and auxiliary-space complexity. (3)
- 10) A list initially contains $10 \rightarrow 20 \rightarrow 30$. Insert 5 at the head, insert 25 after 20, then delete the tail. Draw the final list and state how many next links must be traversed to locate its tail from the head. (2)
- 11) Convert $A+B*(C-D)^E^F$ to postfix using a stack. Exponentiation has highest precedence and is right-associative. Show the operator stack and output after each input token, then give the final postfix expression. (5)
- 12) Write a C function to check whether a string containing only `(, ), [, and ]` is balanced. Use an array stack of capacity 100 and assume the input has at most 100 characters. Return 1 for balanced and 0 otherwise. (3)
- 13) An editor uses undo and redo stacks. Starting with empty stacks, perform edits A, B and C, then undo twice and redo once. List both stacks from bottom to top. State what happens to the redo stack when a new edit D is made. (2)

End of question paper

Worked Solutions

Data Structures

Set B: Original | CSE Semester 3

Companion to ds-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Structure definition

Correct option: (C). `struct` defines a type whose members can hold different kinds of data.

2. Nested member access

Correct option: (A). For structure objects, each member is accessed with a dot: `s.admitted.year`.

3. String capacity

Correct option: (D). One of the 40 positions must hold the terminating null character.

4. Safe input width

Correct option: (B). `%39s` reads at most 39 characters and appends the null terminator.

5. Pointer member access

Correct option: (C). `p->roll` is equivalent to `(*p).roll`.

6. Contiguous dynamic matrix

```
#include <stdlib.h>
int *makeMatrix(int r, int c) {
    if (r <= 0 || c <= 0) return NULL;
    /* The question guarantees that this size fits size_t. */
    return malloc((size_t)r * (size_t)c * sizeof(int));
}
/* Access (i,j) as *(base + (size_t)i * c + j).
Valid indices satisfy 0 <= i < r and 0 <= j < c.
Caller checks the returned pointer and eventually frees it. */
```

Rows are adjacent in memory, so row i starts at offset ic and column j adds j . Allocation requests $rc \text{ sizeof(int)}$ bytes. A non-null result must be initialized before its elements are read and released with `free` when no longer needed.

7. Dangling pointer and memory leak

A dangling pointer refers to storage whose lifetime has ended:

```
int *p = malloc(sizeof *p);
free(p);      /* p now dangles; do not dereference it. */
p = NULL;     /* Clear this pointer after releasing storage. */
```

Any aliases must also stop using that storage. A memory leak loses the final reference to allocated storage:

```
int *p = malloc(sizeof *p);
p = NULL;     /* Wrong: allocated storage can no longer be freed. */
```

Keep an owning pointer and free the allocation before overwriting that final reference. These examples illustrate errors; they are not code to copy into a working implementation.

8. Delete all matching nodes

The pointer link always addresses the pointer that leads to the current node. On deletion, update that pointer and keep link in place so the next node is checked. On retention, advance to the next link. This covers head, consecutive and all-node matches, and releases each removed allocation. Time is $O(n)$; auxiliary space is $O(1)$. The following shared listing also contains the reversal for 9.

```
#include <stdlib.h>
typedef struct Node {
    int value;
    struct Node *next;
} Node;
void deleteAll(Node **head, int key) {
    if (!head) return;
    Node **link = head;
    while (*link) {
        if ((*link)->value == key) {
            Node *old = *link;
            *link = old->next;
            free(old);
        } else link = &(*link)->next;
    }
}
Node *reverse(Node *head) {
    Node *prev = NULL, *cur = head;
    while (cur) {
        Node *next = cur->next;
        cur->next = prev;
        prev = cur;
        cur = next;
    }
    return prev;
}
```

9. Reverse a singly linked list

Use the reverse function in the listing for 8. Save the original successor before changing cur's next pointer; move prev and cur forward. The loop invariant is that prev heads the reversed processed prefix and cur heads the untouched suffix. At termination, prev is the new head. Assign `head = reverse(head)`. Empty and one-node lists work unchanged. Time is $O(n)$ and auxiliary space is $O(1)$.

10. List trace

$$\begin{array}{ccccccc}
 10 \rightarrow 20 \rightarrow 30 & \longrightarrow & 5 \rightarrow 10 \rightarrow 20 \rightarrow 30 \\
 \longrightarrow & 5 \rightarrow 10 \rightarrow 20 \rightarrow 25 \rightarrow 30 & \longrightarrow & \boxed{5 \rightarrow 10 \rightarrow 20 \rightarrow 25}.
 \end{array}$$

Starting at the final head, follow three next links to reach tail 25. Checking that the tail's next pointer is NULL does not traverse another node-to-node link.

11. Stack trace for infix conversion

The stack is listed bottom to top. Exponentiation is right-associative, so the second exponentiation does not pop the first.

Token	Operator stack	Output
A	empty	A
+	+	A
B	+	AB
*	+ *	AB
(	+ * (	AB
C	+ * (	ABC
-	+ * (-	ABC
D	+ * (-	ABCD
)	+ *	ABCD-
^	+ * ^	ABCD-
E	+ * ^	ABCD-E
^	+ * ^ ^	ABCD-E
F	+ * ^ ^	ABCD-EF
End	empty	ABCD-EF^^*+

Thus the final postfix form is `ABCD-EF^^*+`. The two exponentiations encode $(C - D)^{(E^F)}$.

12. Balanced brackets

Push each opening bracket. For a closing bracket, require a nonempty stack and matching opening type. Accept only if the stack is empty at the end. Time is $O(n)$; the fixed array uses capacity 100, as permitted by the question.

```
int balanced(const char *s) {
    char stack[100];
    int top = 0;
    for (; *s; ++s) {
        if (*s == '(' || *s == '[') {
            if (top == 100) return 0;
            stack[top++] = *s;
        } else if (*s == ')' || *s == ']') {
            if (!top) return 0;
            char open = stack[--top];
            if ((*s == ')') && open != '(' ||
                (*s == ']') && open != '[')) return 0;
        } else return 0;
    }
    return top == 0;
}
```

13. Undo/redo trace

Operation	Undo: bottom to top	Redo: bottom to top
Edits A, B, C	A, B, C	empty
Undo	A, B	C
Undo	A	C, B
Redo	A, B	C
New edit D	A, B, D	empty

Undo transfers the newest applied action to redo; redo reapplies its newest undone action. New edit D clears redo, since C belongs to the abandoned editing branch.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A noiseless channel has bandwidth 3 kHz and four signal levels. What is its Nyquist maximum bit rate? (1)
(A) 3 kbps
(B) 6 kbps
(C) 12 kbps
(D) 24 kbps
- 2) How many bits can one of four distinct signal levels represent? (1)
(A) 1
(B) 2
(C) 4
(D) 8
- 3) A channel has bandwidth 3 kHz and linear signal-to-noise ratio 31. What is its Shannon capacity? (1)
(A) 15 kbps
(B) 12 kbps
(C) 93 kbps
(D) 30 kbps
- 4) For the same channel and signalling scheme, Nyquist gives 12 kbps and Shannon gives 15 kbps. Which is the tighter upper bound? (1)
(A) 15 kbps
(B) 27 kbps
(C) 3 kbps
(D) 12 kbps
- 5) Under the noiseless Nyquist model, what happens to the maximum bit rate when bandwidth doubles and the number of levels stays fixed? (1)
(A) It halves
(B) It stays fixed
(C) It doubles
(D) It quadruples

Section B: Theory questions (25 marks)

- 6) Compare ASK, binary FSK and BPSK in terms of the carrier parameter changed and sensitivity to amplitude noise. State one reason to prefer BPSK over ASK in a channel with amplitude disturbances. (3)
- 7) An application generates 900 payload bytes. Each of five layers adds a 20-byte header. Calculate the total transmitted size and the percentage occupied by headers, ignoring trailers. (2)
- 8) Encode data 1101011011 using CRC generator 10011. Show modulo-2 division, the four-bit remainder and the complete transmitted codeword. State the receiver's check for an error-free received word. (5)
- 9) Draw Manchester and bipolar AMI waveforms for 10110010. For Manchester, use low-to-high at mid-bit for 1 and high-to-low for 0. For AMI, assume the last nonzero pulse before this sequence was negative. Label all bit boundaries. (3)
- 10) Explain the different actions of a hub and a learning switch when a frame arrives. For the switch, distinguish known and unknown destination MAC addresses. (2)
- 11) A stop-and-wait link sends 1000-byte data frames at 1 Mbps. One-way propagation delay is 10 ms. ACK transmission and processing times are negligible; no frames are lost. Calculate frame transmission time, time per successful cycle, utilization and useful throughput. Ignore headers. (5)
- 12) A selective-repeat sender sends frames 0, 1, 2 and 3 with window size 4. Frame 1 is lost; frames 0, 2 and 3 arrive and their individual ACKs return before timeout. No new frames are available. State which frames are buffered, which frame is retransmitted and the final in-order delivery sequence. Compare with go-back-N under the same loss, assuming it discards out-of-order frames. (3)
- 13) A shared half-duplex Ethernet link operates at 100 Mbps. The maximum one-way propagation delay is $5\ \mu\text{s}$. Ignoring other delays, calculate the minimum frame length required for collision detection in bits and bytes. (2)

End of question paper

Worked Solutions

Data Communication and Computer Networks

Set B: Original | CSE Semester 3

Companion to dccn-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Nyquist rate

Correct option: (C). $R = 2B \log_2 L = 2(3000) \log_2 4 = 12000$ bps.

2. Bits per symbol

Correct option: (B). Four levels represent $\log_2 4 = 2$ bits per symbol.

3. Shannon capacity

Correct option: (A). $C = B \log_2(1 + S/N) = 3000 \log_2 32 = 15000$ bps.

4. Tighter rate bound

Correct option: (D). Both limits must hold, so the smaller upper bound, 12 kbps, is tighter.

5. Nyquist bandwidth change

Correct option: (C). $R = 2B \log_2 L$ is directly proportional to B when L is fixed.

6. Modulation comparison

ASK changes carrier amplitude, FSK changes carrier frequency, and PSK changes carrier phase. Binary FSK commonly uses two separated frequencies and needs more bandwidth than binary PSK under comparable simple pulse shaping. ASK and BPSK can have comparable nominal bandwidth for the same bit rate and shaping; exact occupied bandwidth depends on the implementation. ASK is directly vulnerable to amplitude disturbances. Constant-envelope FSK and PSK are less affected by such disturbances, although phase/frequency noise and receiver synchronization still matter.

For a coherent receiver in an approximately additive-noise, bandwidth-limited telephone channel, BPSK is a defensible choice: it uses antipodal signals, offers good bit-error performance at a given energy per bit, and avoids the tone separation of binary FSK. A noncoherent, very simple receiver could instead favor FSK. The noise model and implementation must be stated rather than claiming one scheme is always best.

7. Header overhead

Header bytes = $5(20) = 100$. Total bytes = $900 + 100 = 1000$. Header percentage = $100/1000 \times 100 = 10\%$. Payload efficiency is 90%.

8. CRC division and receiver check

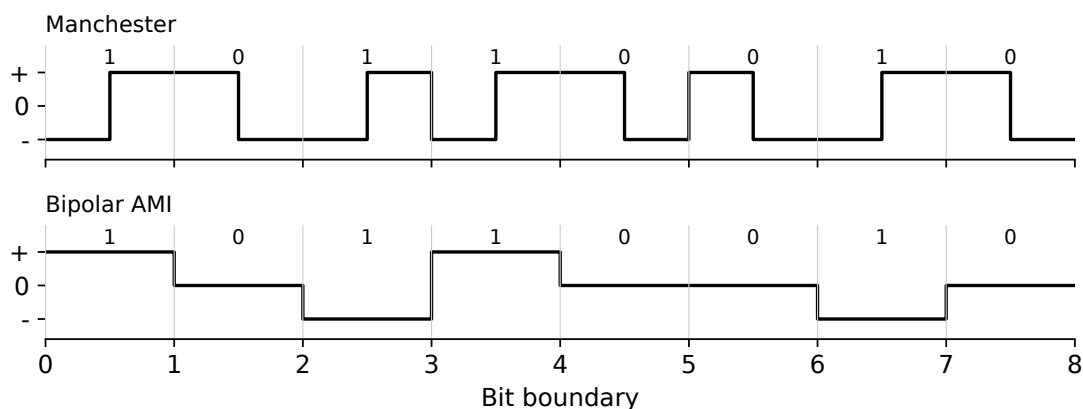
Append four zeros because the generator 10011 has degree four. The aligned XOR working dividends are:

```
11010110110000  initial dividend
01001110110000
00000010110000
00000000101000
00000000001110
```

The remainder is **1110** and the transmitted codeword is **11010110111110**. The receiver divides the whole received codeword by 10011. A zero remainder means no error was detected; it does not guarantee that no undetectable error occurred.

9. Manchester and AMI waveforms

Manchester half-bit pairs are $(-, +), (+, -), (-, +), (-, +), (+, -), (+, -), (-, +), (+, -)$. AMI levels are $+, 0, -, +, 0, 0, -, 0$. These use the conventions printed in the question.



10. Hub and learning switch

A hub repeats the incoming physical signal on every other port. A learning switch records source MAC to ingress-port mappings. For a known destination on a different port, it forwards only there; for a known destination on the incoming port, it filters the frame. For an unknown destination it floods on other forwarding ports in the same VLAN. Broadcasts are also flooded within that VLAN.

11. Stop-and-wait throughput

Frame length $L = 1000(8) = 8000$ bits. Serialization time:

$$t_t = 8000/10^6 = \boxed{8 \text{ ms}}.$$

Cycle time from starting a frame until its ACK returns:

$$t_{\text{cycle}} = t_t + 2t_p = 8 + 20 = \boxed{28 \text{ ms}}.$$

$$U = \frac{8}{28} = \boxed{0.285714}, \quad \text{throughput} = \frac{8000}{0.028} = \boxed{285714.3 \text{ bps}}.$$

Thus utilization is 28.57% and useful throughput is approximately 285.7 kbps. There are no headers or losses in this calculation.

12. Selective repeat versus go-back-N

Selective repeat accepts frame 0 and buffers frames 2 and 3 while waiting for 1. Only frame 1 remains unacknowledged and is retransmitted. After it arrives, the receiver delivers 1, 2 and 3, giving overall delivery order 0, 1, 2, 3. There are five data transmissions in total.

Go-back-N accepts 0, discards 2 and 3, and uses cumulative acknowledgements. After frame 1 times out, it retransmits the outstanding suffix 1, 2 and 3. There are seven data transmissions in total and the same final delivery order.

13. Minimum collision-detection frame

$$L_{\min} = R(2t_p) = 100 \times 10^6 \times 10 \times 10^{-6} = \boxed{1000 \text{ bits} = 125 \text{ bytes}}.$$

This is the propagation-only lower bound for the hypothetical link in the question.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) How many cells are in a four-variable Karnaugh map? (1)
(A) 4
(B) 8
(C) 16
(D) 32
- 2) How many input variables differ between two adjacent Karnaugh-map cells? (1)
(A) 1
(B) 2
(C) 3
(D) 4
- 3) Which is a valid group size when minimizing a sum-of-products expression on a Karnaugh map? (1)
(A) 3
(B) 5
(C) 6
(D) 8
- 4) For $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$, with A most significant, which expression is equivalent? (1)
(A) $B \oplus D$
(B) $\neg B \neg D + BD$
(C) $B + D$
(D) BD
- 5) How can a two-input NAND gate act as an inverter for input X? (1)
(A) Tie both inputs to 0
(B) Leave one input open
(C) Tie both inputs to X
(D) Tie one input to 0

Section B: Theory questions (25 marks)

- 6) Using 5-bit two's complement, evaluate (i) $11 + 7$, (ii) $-9 + 5$, and (iii) $6 - 13$. Give the binary result, its signed interpretation and whether signed overflow occurs in each case. (3)
- 7) Implement $F(A, B, C) = A \oplus B \oplus C$ using a 4-to-1 multiplexer. Choose A and B as select inputs and specify I_0, I_1, I_2, I_3 ; an inverter is available. (2)
- 8) Design a synchronous two-bit counter with T flip-flops and enable input E. For $E = 1$, the state sequence is $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$; for $E = 0$, the state holds. Draw the transition table, derive both T-input equations and draw the circuit. (5)
- 9) Write synthesizable Verilog for a four-bit ring counter with positive-edge clock and active-high synchronous reset. Reset must load 0001; subsequent clocks rotate the one toward the next more significant bit, wrapping from bit 3 to bit 0. (3)
- 10) Explain why an asynchronous ripple counter can briefly produce incorrect decoded outputs. Contrast its clock connection with that of a synchronous counter. (2)
- 11) Design a two-bit unsigned magnitude comparator for $A = a_1a_0$ and $B = b_1b_0$. Derive Boolean expressions for $A > B$, $A = B$ and $A < B$. Draw a gate-level diagram and verify the outputs for $A = 10_2$, $B = 01_2$. (5)
- 12) A byte-addressable memory stores 32-bit words. How many words fit in 4 KiB? How many address bits select individual bytes? If a word starts at hexadecimal address $0x1000$, list the byte addresses it occupies and the starting address of the next word. (3)
- 13) Describe a memory-read operation using the memory address register (MAR), memory data register (MDR) and read control signal. State what changes for a memory-write operation. (2)

End of question paper

Worked Solutions

Digital Systems and Computer Organization

Set B: Original | CSE Semester 3

Companion to dsco-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. K-map cell count

Correct option: (C). There is one cell per input combination, so $2^4 = 16$ cells.

2. K-map adjacency

Correct option: (A). Gray-code ordering makes adjacent cells differ in exactly one variable.

3. SOP grouping

Correct option: (D). Groups contain a power of two cells; 8 is the only power of two listed.

4. Function reduction

Correct option: (B). Every listed minterm has $B = D$. Thus $F = \neg B \neg D + BD$, independent of A and C.

5. NAND inverter

Correct option: (C). Connecting both inputs to X gives $\neg(X \cdot X) = \neg X$.

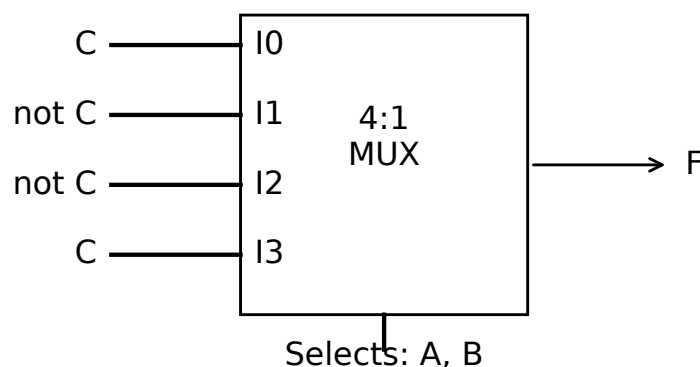
6. Signed results and overflow

Operation	Binary addition	Result	Interpretation	Overflow
11+7	01011 + 00111	10010	-14	Yes
-9+5	10111 + 00101	11100	-4	No
6-13	00110 + 10011	11001	-7	No

Only the true result 18 is outside $[-16, 15]$. For subtraction, 13 is 01101 and its two's-complement negative is 10011. A carry out by itself is not the signed-overflow flag.

7. MUX realization

With A and B as select bits (A most significant), $AB = 00, 01, 10, 11$ produces $C, \bar{C}, \bar{C}, C$, respectively. Thus $I_0 = C, I_1 = \bar{C}, I_2 = \bar{C}, I_3 = C$. One inverter supplies both complemented-C inputs.



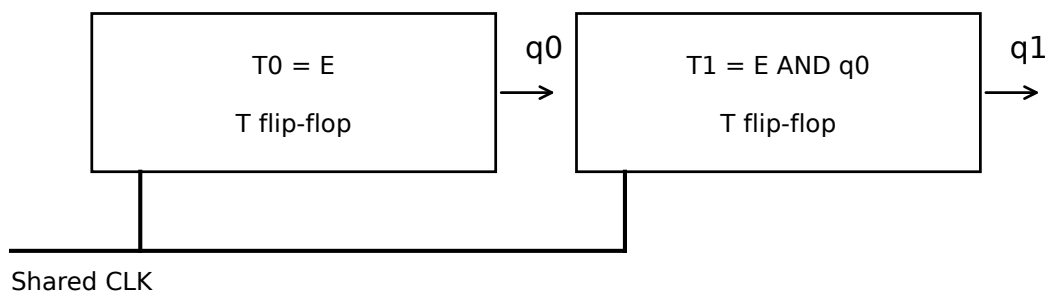
8. Enabled two-bit synchronous counter

Present q_1q_0	Next when E=0	Next when E=1
00	00	01
01	01	10
10	10	11
11	11	00

Use the T excitation equation $T_i = q_i \oplus q_i^+$. The low bit toggles exactly when E=1; the high bit toggles exactly when E=1 and the current low bit is 1.

$$T_0 = E, \quad T_1 = E q_0.$$

Feed E directly to T_0 and an AND of E and q_0 to T_1 . The clock is common.



9. Four-bit ring counter

```

module ring4(input wire clk, input wire reset,
             output reg [3:0] q);
    always @(posedge clk) begin
        if (reset) q <= 4'b0001;
        else q <= {q[2:0], q[3]};
    end
endmodule

```

On a positive clock edge with reset high, q becomes 0001. With reset low, successive states are 0010, 0100, 1000 and 0001. Concatenation moves bits 2:0 upward and puts old bit 3 into bit 0. Nonblocking assignments model sequential storage.

10. Ripple delay and decoded glitches

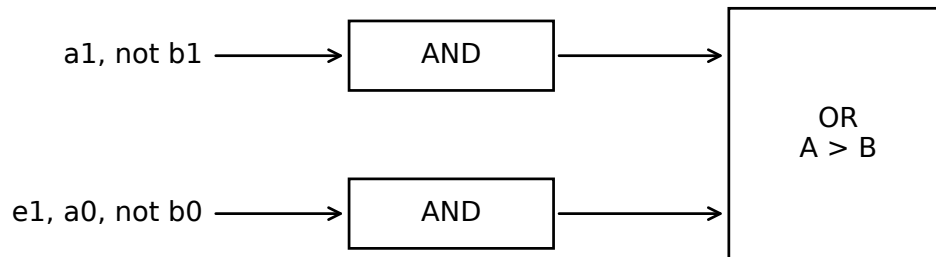
A ripple counter clocks stage 0 from the external clock and later stages from preceding flip-flop outputs. Propagation delays accumulate, so several output bits do not change simultaneously. During a transition such as 0111 to 1000, a decoder may briefly recognize an intermediate pattern. A synchronous counter drives all flip-flops from one shared clock and computes excitation logic from the old state. It avoids stage-by-stage clock ripple, although real output skew and combinational delays still require proper timing design.

11. Two-bit magnitude comparator

Let $e_i = a_i \text{ XNOR } b_i$. Then

$$G = a_1 \bar{b}_1 + e_1 a_0 \bar{b}_0, \quad E_q = e_1 e_0, \quad L = \bar{a}_1 b_1 + e_1 \bar{a}_0 b_0.$$

The upper bit decides unless equal; only then does the lower bit decide. Use XNOR gates for e_1, e_0 , inverters for complements, AND gates for product terms and OR gates to combine them. The figure gives the G network. Swap a and b for L, and AND the two XNOR outputs for E_q .



For A=10 and B=01, $e_1 = 0$, $a_1 \bar{b}_1 = 1$, so $G = 1, E_q = 0, L = 0$.

12. Memory capacity and addresses

4 KiB = 4096 bytes. Each 32-bit word occupies 4 bytes:

$$\text{words} = 4096/4 = 1024, \quad \text{byte-select bits} = \log_2(4096) = 12.$$

The word at 0x1000 occupies 0x1000, 0x1001, 0x1002 and 0x1003. The next word starts at 0x1004. The 12 bits select an offset within the 4 KiB block; the full system may use more address bits to place that block at base 0x1000. Endianness changes byte significance, not these occupied addresses.

13. Memory read and write

For a read: place the target address in MAR, assert the read control signal, wait until the memory interface indicates valid data, and latch the returned value into MDR. Transfer MDR to the destination register as required.

For a write: place the target address in MAR and outgoing data in MDR, assert write with the required timing, and wait for completion. Read moves data from memory to MDR; write moves data from MDR to memory. MAR selects the location in both cases.

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the least element? (1)
(A) 0
(B) 2
(C) 1
(D) 12
- 2) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the greatest element? (1)
(A) 1
(B) 12
(C) 6
(D) 4
- 3) In the divisor lattice D_{12} , what is $4 \wedge 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 4) In the divisor lattice D_{12} , what is $4 \vee 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 5) Which is a complement of 2 in D_{12} , with meet given by gcd and join by lcm? (1)
(A) 3
(B) 6
(C) No element
(D) 12

Section B: Theory questions (25 marks)

- 6) For $F(x, y, z) = (x \wedge \neg y) \vee z$, construct the truth table and write the canonical disjunctive normal form and canonical conjunctive normal form. Use variable order (x, y, z) . (3)
- 7) Prove the Boolean identity $a \vee (\neg a \wedge b) = a \vee b$ using distributive and complement laws. Name the laws used. (2)
- 8) Count the integer solutions of $x + y + z = 12$ subject to $1 \leq x \leq 5$, $2 \leq y \leq 6$ and $z \geq 0$. Transform to nonnegative variables and use inclusion-exclusion. Show each excluded case. (5)
- 9) Find the 42nd lexicographic permutation of the symbols 0, 1, 2, 3, 4, with 01234 counted as the first. Show the factorial-block choices. (3)
- 10) How many distinct arrangements of the letters of BALLOON are possible? How many have the two Ls adjacent? Explain how repeated letters are handled. (2)
- 11) Use Dijkstra's algorithm from A in an undirected weighted graph with vertices A, B, C, D, E and edges AB:4, AC:1, BC:2, BD:1, CD:5, CE:8, DE:3. Show the tentative-distance table after settling each vertex. Give a shortest path and distance from A to every other vertex. (5)
- 12) Prove that a simple graph with $n \geq 2$ vertices and minimum degree at least $(n - 1)/2$ is connected. Hint: bound the size of each component if the graph were disconnected. (3)
- 13) A simple graph is regular and self-complementary on n vertices. Use degree counting to show its degree is $(n - 1)/2$ and then show that $n \equiv 1 \pmod{4}$. (2)

End of question paper

Worked Solutions

Discrete Mathematical Structures

Set B: Original | CSE Semester 3

Companion to dms-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Bottom of divisor lattice

Correct option: (C). 1 divides every element of this set.

2. Top of divisor lattice

Correct option: (B). Every element of the set divides 12.

3. Meet of divisors

Correct option: (A). The meet is the greatest common divisor, $\gcd(4, 6) = 2$.

4. Join of divisors

Correct option: (D). The join is the least common multiple, $\text{lcm}(4, 6) = 12$.

5. Complement existence

Correct option: (C). A complement c needs $\gcd(2, c) = 1$ and $\text{lcm}(2, c) = 12$. The only odd divisors, 1 and 3, give lcms 2 and 6; neither works.

6. Truth table and canonical forms

x	y	z	$\neg y$	$x \wedge \neg y$	F
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

Thus $F = \Sigma m(1, 3, 4, 5, 7) = \Pi M(0, 2, 6)$.

Canonical DNF: $F = (\neg x \wedge \neg y \wedge z) \vee (\neg x \wedge y \wedge z) \vee (x \wedge \neg y \wedge \neg z) \vee (x \wedge \neg y \wedge z) \vee (x \wedge y \wedge z)$.

Canonical CNF: $F = (x \vee y \vee z) \wedge (x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee z)$.

7. Boolean identity proof

$$\begin{aligned}
 a \vee (\neg a \wedge b) &= (a \vee \neg a) \wedge (a \vee b) && \text{distributive law} \\
 &= 1 \wedge (a \vee b) && \text{complement law} \\
 &= a \vee b && \text{identity law.}
 \end{aligned}$$

8. Integer solutions by inclusion-exclusion

Set $u = x - 1$ and $v = y - 2$. Then $u + v + z = 9$, with $0 \leq u, v \leq 4$ and $z \geq 0$. The unrestricted count is $\binom{11}{2} = 55$. Violations $u \geq 5$ and $v \geq 5$ each leave a nonnegative sum of 4 and each contribute $\binom{6}{2} = 15$. Both cannot occur simultaneously because their sum would be at least 10. Therefore

$$55 - 15 - 15 = 25.$$

As a check, each of the five possible x values can pair with each of the five possible y values; even the maximum sum is $5 + 6 = 11 \leq 12$, and each pair fixes a nonnegative z .

9. 42nd lexicographic permutation

Use zero-based rank $42 - 1 = 41$. Factorial block choices are:

Available symbols	Block size	Quotient / remainder	Chosen
0,1,2,3,4	24	$41 = 1(24) + 17$	1
0,2,3,4	6	$17 = 2(6) + 5$	3
0,2,4	2	$5 = 2(2) + 1$	4
0,2	1	$1 = 1(1) + 0$	2
0	1	$0 = 0(1) + 0$	0

Choose the quotient-indexed symbol from the sorted remaining list at each step, with zero-based indices. The result is $\boxed{13420}$.

10. Arrangements of BALLOON

There are 7 letters, with L repeated twice and O repeated twice. The number of distinct arrangements is

$$\frac{7!}{2!2!} = 1260.$$

If the two Ls must be adjacent, treat LL as one object. The six objects are LL, B, A, O, O, N, so the count is

$$\frac{6!}{2!} = 360.$$

There is no additional factor for exchanging the identical Ls within their block.

11. Dijkstra on the undirected graph

Initialize A to 0 and every other tentative distance to infinity. All weights are nonnegative. After each settlement, relax each incident edge.

Settled	A	B	C	D	E
Initial	0	∞	∞	∞	∞
A	0	4	1	∞	∞
C	0	3	1	6	9
B	0	3	1	4	9
D	0	3	1	4	7
E	0	3	1	4	7

Destination	Shortest path	Distance
A	A	0
B	$A \rightarrow C \rightarrow B$	3
C	$A \rightarrow C$	1
D	$A \rightarrow C \rightarrow B \rightarrow D$	4
E	$A \rightarrow C \rightarrow B \rightarrow D \rightarrow E$	7

12. Connectivity proof

Suppose the graph is disconnected. If a component has s vertices, every vertex in it has degree at most $s - 1$. The minimum-degree assumption forces

$$s \geq \delta(G) + 1 \geq (n + 1)/2.$$

There are at least two components, so their combined size would be at least $n + 1$, contradicting the total of n vertices. Hence the graph is connected.

13. Degree and congruence

For a nonempty simple r -regular graph, its complement is $(n - 1 - r)$ -regular. Self-complementarity gives $r = n - 1 - r$, hence $r = (n - 1)/2$. Thus n is odd. By the handshake lemma, nr is even, so r must be even because n is odd. Writing $r = 2t$ gives $n = 4t + 1$, or $n \equiv 1 \pmod{4}$.